

Features of the I2C peripheral:

- I²C-bus specification and user manual rev. 5 compatibility:
 - Target and controller modes, multicontroller capability
 - Standard-mode (Sm), with a bitrate up to 100 kbit/s
 - Fast-mode (Fm), with a bitrate up to 400 kbit/s
 - Fast-mode Plus (Fm+), with a bitrate up to 1 Mbit/s and extra output drive I/Os
 - 7-bit and 10-bit addressing mode, multiple 7-bit target addresses
 - Programmable setup and hold times
 - Clock stretching
- SMBus specification rev 3.0 compatibility:
 - Hardware PEC (packet error checking) generation and verification with ACK control
 - Command and data acknowledge control
 - Address resolution protocol (ARP) support
 - Host and device support
 - SMBus alert
 - Timeouts and idle condition detection
- PMBus rev 1.3 standard compatibility
- Independent clock: a choice of independent clock sources allowing the I²C-bus communication speed to be independent of the PCLK reprogramming
- Wake-up from Stop mode on address match
- Programmable analog and digital noise filters
- 1-byte buffer with DMA capability

Table 8. I²C implementation

I ² C features ⁽¹⁾	I2C1	I2C2
Standard mode (up to 100 kbit/s)	X	X
Fast mode (up to 400 kbit/s)	X	X
Fast Mode Plus (up to 1 Mbit/s) with extra output drive I/Os	X	X
Programmable analog and digital noise filters	X	X
SMBus/PMBus hardware support	X	-
Independent clock	X	-
Wakeup from Stop mode on address match	X	-

1. X: supported

3.18 Universal synchronous/asynchronous receiver transmitter (USART)

The devices embed two universal synchronous/asynchronous receivers/transmitters that communicate at speeds of up to 6 Mbit/s.

They provide hardware management of the CTS, RTS and RS485 DE signals, multiprocessor communication mode, synchronous SPI communication and single-wire half-